

An Off-chip Capacitor-free Low-Dropout Regulator with Enhanced Slew-Rate

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Abstract—The fully integrated low-dropout regulator (LDO) without external capacitors provides a low-noise, low-ripple power supply voltage suitable for highly integrated system-on-chip (SoC) applications. To address the issues of slow transient response and reduced stability associated with off-chip capacitor-free LDOs, a novel transient enhancement circuit is proposed to accelerate the charging and discharging rates of the power transistor, thereby improving the LDO's transient response speed. This design is based on a standard 180 nm CMOS process and employs a 2 pF Miller compensation capacitor for stabilization. It can deliver a maximum load current of 20 mA with a voltage drop of 200 mV over a supply voltage range of 3 V to 1.2 V. The quiescent current of the LDO is 26 μ A under no-load conditions. When the load current switches between 100 μ A and 20 mA with a rise and fall time of 50 ns, the overshoot voltage is 71 mV with a recovery time of 0.32 μ s, while the undershoot voltage is 88 mV with a recovery time of 0.44 μ s.

Keywords—Capacitor-free LDO, Transient enhanced, Miller compensation.

I. INTRODUCTION

The low-dropout regulator is an essential and fundamental component of integrated power management systems. Conventional LDOs with off-chip capacitor typically require a μ F-level external large capacitance, which entails considerable area and power consumption. For noise and power sensitive applications such as the Internet of Things (IoT) [1], low-power, capacitor-free LDOs have been widely adopted. For fully integrated LDOs, three critical aspects that are currently the focus of extensive research include load transient response, loop stability, and power supply rejection ratio (PSRR) [2]. These metrics are important for robust SoC power delivery. Fast load transient response is critical to prevent significant output voltage droop or overshoot during

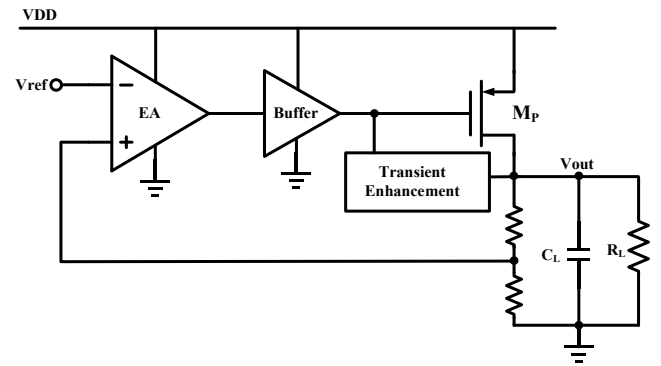


Fig. 1. Block diagram of the proposed Capacitor-free LDO.

rapid current demand changes from digital circuits, which could disrupt noise sensitive analog/RF blocks [3], [4]. Loop stability must be guaranteed across all operating conditions to avoid harmful oscillations that degrade performance or cause system failure. High PSRR is essential to attenuate supply-borne noise originating from the digital domain or other switching regulators sharing the same supply rail. Achieving high performance in all three aspects simultaneously is particularly challenging in off-chip capacitor-free LDO designs. The absence of a large external capacitor, while beneficial for full integration and reduced board area, removes a major stabilizing element and energy reservoir, exacerbating transient voltage spikes and complicating frequency compensation [5], [6]. Consequently, innovative circuit techniques are demanded to meet stringent specifications within the constraints of capacitor-free LDO for modern SoC applications.

To address the above challenges, researchers have proposed a series of solutions. Cai introduces an LDO architecture based on a flipped voltage follower (FVF) [7],

This work is supported by National Natural Science Foundation of China with project number 62174181.

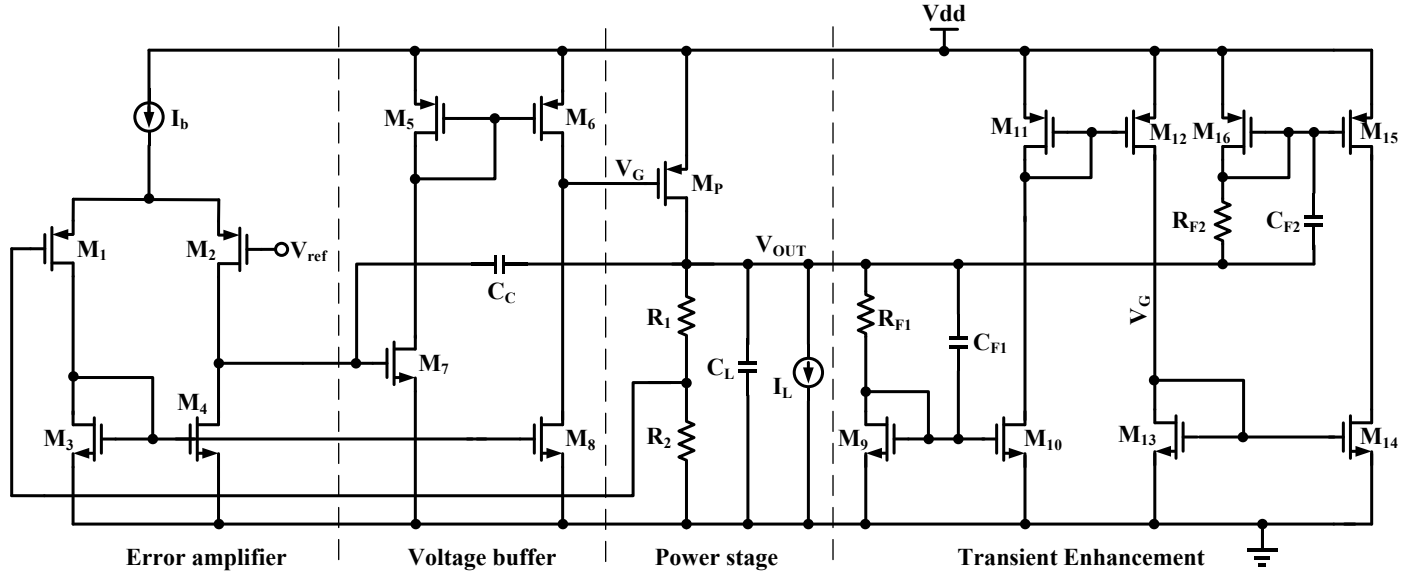


Fig. 2. Full schematic of the proposed LDO.

which exhibits high loop stability and rapid transient response. However, due to the absence of an error amplifier in the feedback loop, the loop gain is relatively low, resulting in higher load regulation and line regulation. Chen designs an additional transient enhancement circuit to improve the transient response of the LDO [8]. Nonetheless, this approach may adversely affect loop stability and consumes additional energy. Manda incorporates a multi-loop FVF structure to enhance load regulation [9]. However, the increased number of loops limits stability and significantly raises power consumption and circuit complexity.

Therefore, this paper proposes an off-chip capacitor-free LDO with a novel transient enhancement circuit, as shown in Fig. 1. The circuit utilizes a single-stage amplifier combined with Miller compensation to set the error amplifier's output as the dominant pole of the system, thereby enhancing loop stability. The second stage not only provides strong drive capability but also increases the loop gain of the LDO, thereby reducing both load regulation and line regulation. Additionally, a new transient enhancement circuit is integrated to improve the transient response, resulting in reduced overshoot and undershoot voltages. The organization of this paper is as follows: Section II introduces the design of the proposed LDO. Section III illustrates the simulation results of the proposed design. Conclusions are drawn in Section IV.

II. DESIGN OF THE PROPOSED LDO

The complete schematic of the proposed capacitor-free LDO is shown in Fig. 2. It consists of $M_1 \sim M_4$ forming a single-stage OTA amplifier, while $M_5 \sim M_8$ constitute a push-pull output stage with strong drive capability, capable of rapidly charging and discharging the power transistor M_P . The power stage of the circuit comprises the power transistor M_P , voltage divider resistors R_1 and R_2 , and the on-chip output capacitor C_L . C_C is the Miller compensation capacitor, which serves to split the poles and ensure that the error

amplifier's output acts as the dominant pole. $M_9 \sim M_{16}$, along with R_{F1} , R_{F2} , C_{F1} , and C_{F2} , form the transient enhancement circuit, including overshoot and undershoot suppression circuits.

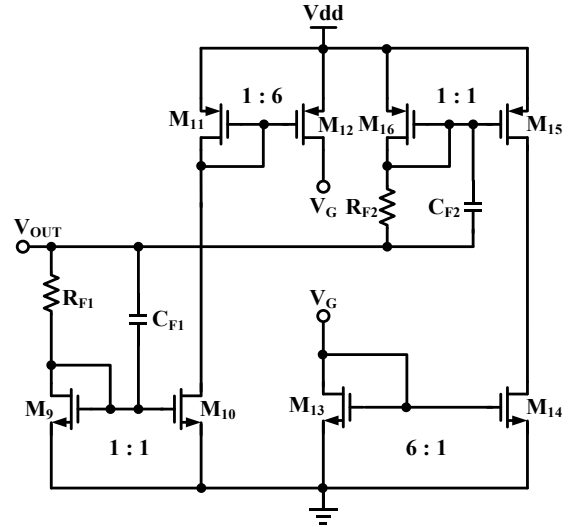


Fig. 3. The schematic of the transient enhancement circuit.

The transient enhancement circuit is shown in Fig. 3, where $M_9 \sim M_{12}$, R_{F1} , and C_{F1} form the overshoot suppression circuit, and $M_{13} \sim M_{16}$, R_{F2} and C_{F2} constitute the undershoot suppression circuit. R_{F1} and R_{F2} are used to adjust the static operating points of the gates of M_9 and M_{16} , respectively, ensuring that M_9 and M_{16} can quickly conduct when large signal fluctuations occur at the output, while also preventing false conduction caused by noise or other factors.

When the load current of the LDO remains constant, $M_9 \sim M_{16}$ are biased in the subthreshold region. Coupling capacitors C_{F1} and C_{F2} are used to couple the overshoot and undershoot output voltage to the gates of M_{10} and M_{15} . When

the load current jumps from heavy to light load, the gate voltage of M_P cannot rise immediately, causing the load current to charge the output capacitor C_L , resulting in an overshoot peak at the output voltage V_{OUT} . This overshoot peak is coupled via C_{F1} to the gate of M_{10} , causing a transient increase in the current through M_{10} . This current is then amplified by the current mirrors (M_{11} and M_{12}), which provide an additional charging path to the gate capacitor of M_P , accelerating the increase of the M_P gate voltage and thus reducing the overshoot voltage and recovery time. To achieve low power consumption while allowing a larger gate charging current for M_P , the W/L ratios of M_9 and M_{10} are set to 1:1, and those of M_{11} and M_{12} are set to 1:6.

Similarly, when the load current transitions from light to heavy load, the gate voltage of M_P cannot decrease immediately, preventing M_P from supplying the load current promptly and causing a undershoot peak at V_{OUT} . This undershoot peak is coupled through C_{F2} to the gate of M_{15} , resulting in a transient increase in the current through M_{15} . The current is then amplified by the current mirrors (M_{13} and M_{14}), which provide an additional discharging path for the gate capacitor of M_P , accelerating the decrease of the M_P gate voltage and thus reducing undershoot voltage and recovery time. Similarly, the W/L ratios of M_{16} and M_{15} are set to 1:1, and those of M_{14} and M_{13} are set to 1:6.

III. SIMULATION RESULTS OF THE PROPOSED LDO

The proposed capacitor-free LDO is designed in a standard 180 nm CMOS process, which input voltage can be 1.2V – 3V and regulated output voltage is 1.0V. The LDO supports a maximum load current of 20mA while consuming only 26 μ A quiescent current I_Q . The total on chip capacitors including C_C , C_{F1} and C_{F2} are 2.4pF.

A. Stability

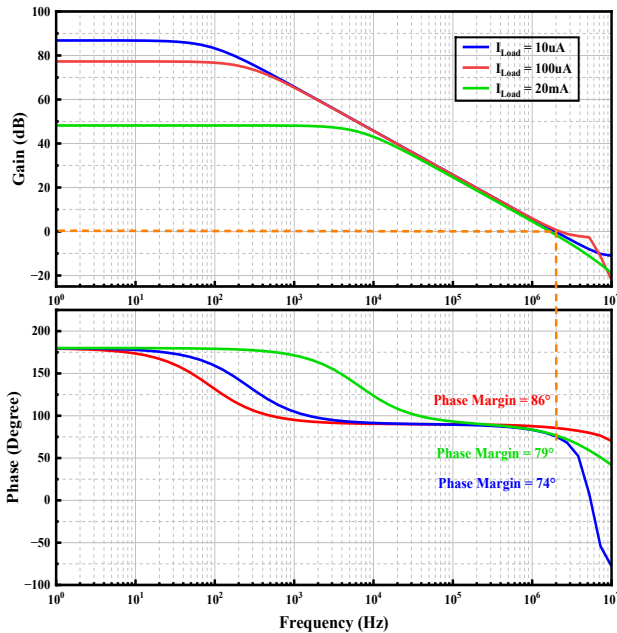


Fig. 4. Bode plots of the proposed LDO with different I_{Load} .

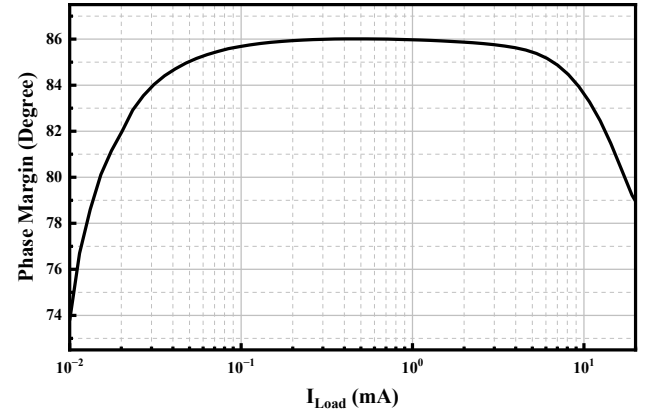


Fig. 5. Phase margin of the proposed LDO under 10 μ A -20mA I_{Load} .

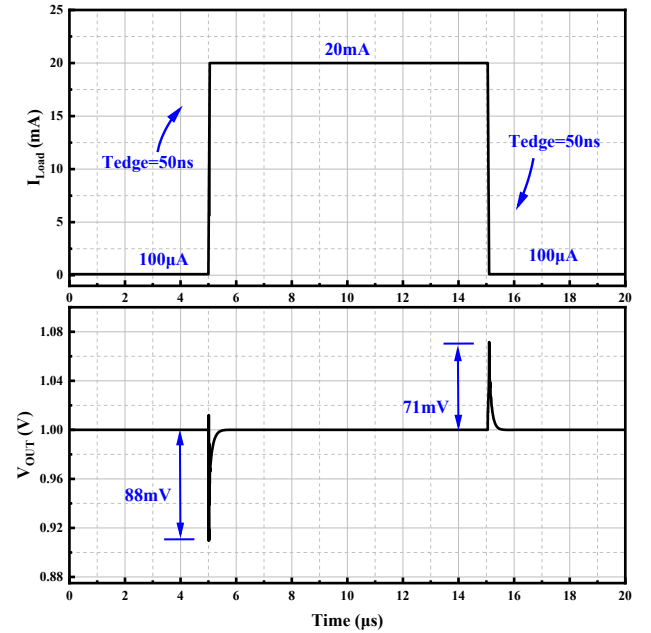


Fig. 6. Simulated load transient response with $C_L = 50$ pF.

The proposed LDO supports a minimum output current of 10 μ A, corresponding to an load current range of 10 μ A to 20 mA. The simulated loop gain and phase margin of the proposed LDO with $C_L = 50$ pF at different I_{Load} are shown in Fig. 4. The loop gain and frequency response of the LDO vary with the load current. When the load current is large, the loop gain is reduced, resulting in better stability. Conversely, at lower load currents, the loop gain increases, and stability diminishes. Under light load conditions with $I_{Load} = 10\mu$ A, the low frequency gain is 86 dB, the bandwidth is approximately 2 MHz, and the phase margin is 74°. Under heavy load conditions with $I_{Load} = 20$ mA, the low-frequency gain is 48 dB, and the phase margin is 79°.

Additionally, Fig. 5 illustrates the phase margin of the proposed LDO under a load current range of 10 μ A to 20 mA. It can be observed that the minimum phase margin across the entire load current range is 74°. This further validates that the

TABLE I. PERFORMANCE COMPARISONS

Publication	2022[10]	2023[11]	2023[12]	2023[13]	2024[14]	2025[15]	This Work
Tech.(nm)	180	180	180	180	180	180	180
$V_{IN}(V)$	1.8	1.2-1.8	1.1-2.5	2.2-3.4	1.65-1.8	1.15	1.2-3
Dropout(mV)	200	200	100	400	150	150	200
$V_{OUT}(V)$	1.6	1.0	1.0	1.8	1.5	1.0	1.0
$C_L(pF)$	100	0-100	100	260	0-100	50	50
$I_Q(\mu A)$	25.3	79	1.8	220-308	24	6	26
$I_{MAX}(mA)$	100	100	10	100	20	30	20
$I_{MIN}(\mu A)$	200	100	100	0	100	0	100
$\Delta V_{OUT}(mV)$	126	253	140	184	172	200	71
Tedge(ns)	300	100	500	100	10	100	50
Tsettle(μs)	0.3	0.34	0.2	0.81	0.02	0.1	0.32

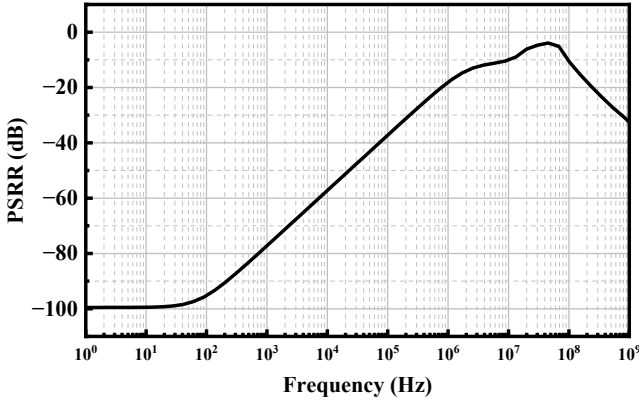


Fig. 7. Simulation result of PSRR for the proposed LDO.

proposed LDO maintains stability across the entire load current range.

B. Load Transient

In the simulations, this paper assumes 100 μA is the minimum load transient current. LDO load transient response for transition between 100 μA and 20mA for the proposed design is shown in Fig. 6. Simulated rise and fall load time is 50ns. When the load current switches from light load to heavy load, the undershoot voltage is 88 mV, with a recovery time of 0.44 μs . Conversely, when switching from heavy load to light load, the overshoot voltage is 71 mV, and the recovery time is 0.32 μs .

C. PSRR

In addition to transient response capability, the ability of the LDO to suppress power supply noise is a key performance metric. Fig. 7 shows the simulated Power Supply Rejection

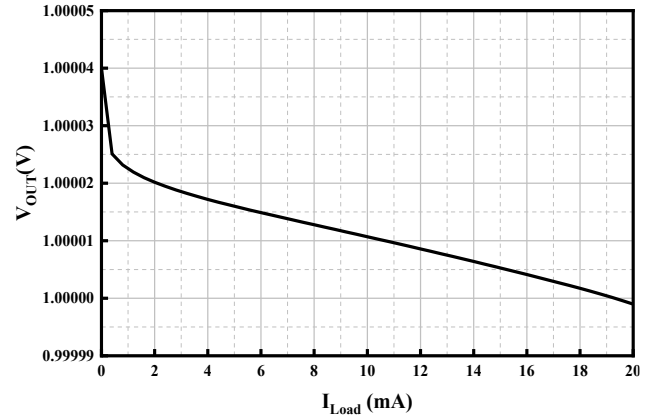


Fig. 8. Simulated load regulations of the proposed LDO.

Ratio of the proposed design at 1.2V supply voltage and 1.0V output voltage. The load current is 1mA and a 50pF load capacitor is used. A PSRR of -56dB is observed at 10kHz.

D. Load Regulation

Fig. 8 shows the load regulation of the proposed LDO at $V_{in} = 1.2 V$ for a load current from 10 μA to 20 mA, with a regulation of 0.002 mV/mA.

Table 1 presents performance comparisons between the proposed LDO and recently published works, which shows that the LDO proposed in this paper has better performance.

IV. CONCLUSION

This paper discusses a 26 μA I_Q 1.0V off-chip capacitor-free LDO with enhanced transient performance via enhanced slew rate capable of delivering up to 20mA load current. By integrating the overshoot suppression circuit and the undershoot suppression circuit, the proposed design

effectively reduces the transient overshoot and undershoot caused by large load current jumps, while also shortening the recovery time. Simulation results indicate that the LDO maintains sufficient gain and phase margin under both light and heavy load conditions. When the load current switches between 100 μ A and 20 mA with a rising and falling edge of 50 ns, the overshoot voltage of the LDO is 71 mV with a response time of 0.32 μ s, while the undershoot voltage is 88 mV with a response time of 0.44 μ s. The proposed structure has a total on-chip capacitance of 2.4pF, which is suitable for highly integrated low voltage SoC applications.

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